

DesertWoZ: A Wizard of Oz Environment to Support the Design of Collaborative Conversational Agents

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ABSTRACT

The use of conversational agents to support team-based training in virtual environments is a nascent area of research within the Computer Supported Collaborative Work (CSCW) and Conversational User Interface (CUI) communities. To support this research, we developed a Wizard of Oz (WoZ) system enabling experimenters to play the role of the conversational agent who must work with a team of three human participants to herd together a group of autonomous robots (target agents) within a virtual task environment. The experimenter can select from one of several tailored responses. The aim of this environment is to determine whether the identified responses facilitate effective team performance and to collect training data to build a conversational agent capable of engaging in team based virtual learning environments.

CCS CONCEPTS

• **Computing Methodologies** → Artificial Intelligence; Natural Language Processing; • **Applied Computing** → Education; Collaborative Learning; • **Human-centered Computing** → Collaborative and Social Computing.

KEYWORDS

virtual environment, conversational user interface, collaborative agent, wizard of oz

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1 INTRODUCTION

Technology is increasingly being used to build virtual environments that support a wide variety of training objectives from military exercises [4] to preparing for a challenging job interview [15]. The benefits of such environments include ability to train in environments that would either be dangerous, expensive, or logistically difficult to perform in the real-world. Such environments also provide educators with the ease and flexibility to customize training scenarios for individual learning objectives. Indeed, virtual collaborative environments have seen a significant uptick in interest in recent times due to the ongoing COVID-19 pandemic [17]. Despite its many benefits, one challenge unique to employing virtual environments for team training is organizing virtual training exercises, particularly when teams are physically and/or temporally dispersed. For instance, experts who play a crucial role in participating in such training exercises may also be unavailable, resulting in less-than-ideal training exercises. One potential solution to this problem is the use of Artificial Intelligence (AI) agents that can assume the role of missing participants. Such agents offer several benefits including their ability to constantly learn new strategies, their patience in working with humans who may struggle to learn a particular topic, and the possibility for such agents to adapt to participant's learning needs and styles. One particular use of AI agents to support team training in virtual environments is the use of Conversational User Interfaces (CUIs) capable of listening and responding to participants in the virtual environment [11].

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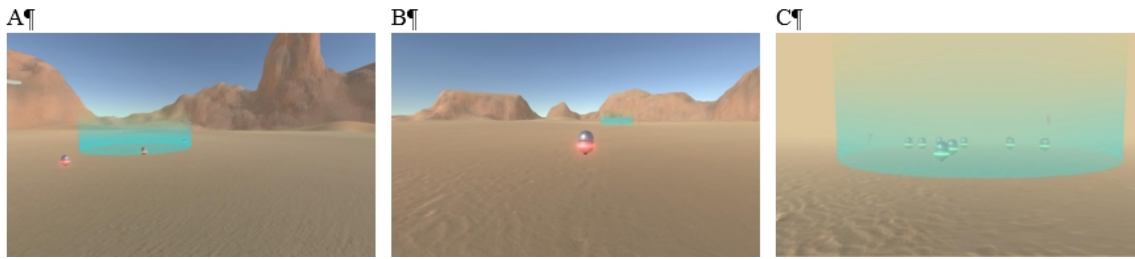


Figure 1: DesertWoZ environment. (a) ground player's view in a clear condition; (b) ground player chasing an autonomous agent towards the containment area; (c) ground player's view in the foggy condition

Conversational User Interfaces (CUIs) have grown in popularity in recent years due in large part to the significant investments made by Amazon, Google, and other major companies into the technology including on their own consumer conversational platforms. For instance, Amazon has shipped over 67 million Echo smart speakers which feature its Alexa conversational agent since 2020 [18]. Indeed, Amazon boasts Alexa now supports over 100,000 different voice commands developed by third parties[7]. Relatedly, there has been growing interest in the industry and research community to develop CUIs to support a variety of capabilities from healthcare [9, 12] to museum exhibitions [2].

A popular method CUIs designers often employ to prototype and evaluate CUIs are Wizard of Oz (WoZ) studies[5] where participants are told they are interacting with an autonomous conversational agent when in reality, the agent is being controlled by the experimenter. This methodology enables designers to rapidly prototype and evaluate how the conversational agent could function without having to dedicate significant resources towards building a full-fledged one. Over the years, the research community has developed several tools to facilitate the prototyping and evaluation of CUIs through via WoZ [8, 14]. However, these tools are designed to prototype stand-alone CUIs whose interaction is constrained to the predominant single user model[10]. To address these shortcomings, we developed DesertWoZ, a Wizard of Oz interface that enables researchers to study how CUIs engaged in a virtual environment could support teams of 3 human participants engaged in a learning activity.

2 THE DESERTWOZ ENVIRONMENT

To examine the use of CUIs in a collaborative virtual learning environment, we extend a previously described virtual learning environment [13]. In this virtual desert environment, teams of four players are tasked with locating and ultimately containing autonomous agents into a containment zone (Figure 1). Three of the players are human participants given the role of a ground-player who are in the desert environment and can directly locate and corral the autonomous targets into the containment area (Figures 1). Each ground player is given a unique color which can be used by team members to identify them. The fourth participant was given the role of operator who had a birds-eye-view of the environment but cannot directly interact with the autonomous agents (Figure 2). The only way the Operator can influence the outcome of the trial is by verbally communicating what they see on their birds-eye-view of

the environment, answering enquiries from ground players, and ultimately requesting ground players to perform specific actions in the environment. Participants are told the operator they are interacting with is an AI agent and that voice chat would be used to communicate with it. However, the operator agent is controlled by the experimenter who selects the response the operator agent makes. Participants are given a time limit (e.g. 5 minutes) to collect and corral the numerous (e.g. 9) autonomous agents on the field. In addition, the visibility of the environment can be adjusted to be either clear (Figure 1), or a foggy (Figure 1) environment.

2.1 The Operator Agent WoZ Interface

The experimenter playing the role of the Operator CUI agent in the study is presented with a WoZ interface (Figure 2). In the center of this interface is a map view which displays the position of each ground player, target agents ground players need to collect and the containment area where the target agents need to be corralled into. On the right-hand side of the interface, there is a panel of buttons, dropdown menus and text fields the Wizard will use to select and customize the response they want to provide to the ground players. On the lower left side of the interface, there is a panel which shows what the Wizard has already said and when, enabling the Wizard to confirm responses were made.

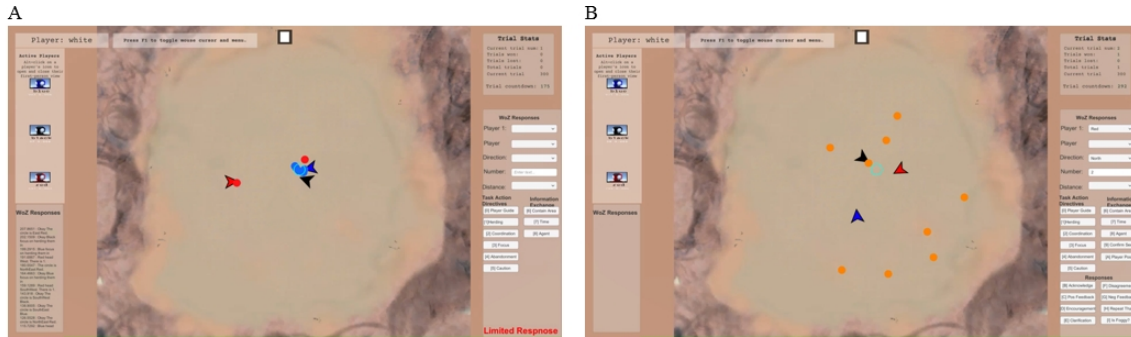
The Wizard has a total of 19 types of responses they can choose from (Table 1). These response types were identified based on a close analysis of the top responses made by human participants who played the role of Operator in a prior version of this study[13], along with prior literature that sought to categorize similar responses (e.g.,[3, 6, 16]). These responses can be broadly classified into three high categories: commands given to ground players (Task Action Directives), sharing of environmental information (Information Exchange), and simple statements and close-ended responses (Responses). The Wizard can only select from these pre-defined responses as we are interested in understanding whether the envisioned functionality of an Operator CUI can facilitate effective team performance and communication in this environment.

To tailor the individual responses to a given circumstance, the Wizard has access to one of several, “slots” they can fill in (Table 2). For instance, selecting the response Player Guidance, with the slots Primary 1: Red, Direction: East, and Number set to 2 would cause the wizard to utter, “Red head east. There are two there”. To aid the Wizard in performing the task, keyboard shortcuts have been mapped to each response and commonly used slots.

Table 1: Supported Responses from Operator Wizard

Domain	Intent	Description	Example Response
Task Action Directive	Player Guidance	Order player where to go	Red head north a little, there are three there
	Herding	Command player to focus on herding	Blue focus on herding those in
	Instruction		
	Coordination	Command player to focus on existing task	Red help blue, they're north from you
	Abandonment	Command player to abandon task	Black can you focus on those for now?
Information Exchange	Caution	Command player to exercise caution	Just leave those, blue.
	Containment	Advise player on where containment area is for them	The circle is to your right red.
	Area		
	Time	Advise players on amount of time remaining in trial	We've got 30 seconds remaining
	Agent	Advise players on number of agents on field.	There are 9 bots.
Responses	PlayerPos*	Advise players of location of other players	Blue is north of you, red.
	Acknowledgement*	Confirm requests, response, and commands have been received	Okay
	Disagreement*	Disagree to proposed task/action/plan	No
	Clarification*	Clarify who the operator is referring to	I'm referring to red player
	Positive Feedback*	Provide positive feedback to player/team	Good work, team!
	Negative Feedback*	Provide negative feedback to player/team	Bad work, team!
	Encouragement*	Provide encouragement to player/team	Don't worry, we'll do better next time!
	Confirm See*	Determine whether player can see what you see	Can you see that, red?
	Repeat*	Ask player to repeat what was said	Could you repeat that, red?
	Foggy*	Confirm whether Desert environment is foggy	Is it foggy out there?
	Wrong Way	Advise player they are going the wrong way	You are going the wrong way, blue
	Right Way	Advise player they are going the right way	You are going the right way, red

* Intent Only Supported in Full Response Mode

**Figure 2: WoZ interface in its two modes: (a) Limited Response; (b) Full Response.****Table 2: Supported Slots**

Slot Name	Values
Player 1, Player 2	Red, Black, Blue
Direction	North, East, West, South, Northeast, Northwest, Southeast, Southwest
Number	Numeric value between 0 and 300
Distance	Very Far Away, Far Away, Close, Very Close

The WoZ interface supports two types of Wizard configurations. The first configuration, referred to Limited Response mode (Figure 2) constrains the Operator's available utterances to only those which do not respond to a ground player's enquiry or would require a response from the ground player being addressed. The Wizard also cannot listen to what ground players are saying, in this condition, in essence blindly providing commands and information as they see fit. A bold red label appears at the bottom right corner of the screen in this mode, reminding the Wizard that they are in the Limited Response mode. In the second configuration, referred to as Full Response mode (Figure 2), the Wizard's available utterances grows to the complete 20 utterances with the Wizard being able to listen to the team communication so that it could better tailor its utterances.

3 TECHNICAL DESCRIPTION

The DesertWoZ environment was built using Unity¹ and uses a client/server architecture. The server interface enables the experimenter to not only control the experiment but also provides the aforementioned WoZ interface. Participants run a client executable on their computer which connects to the server and allows them to join as a ground player. All participants, the experimenter, and the Wizard agent (controlled by the experimenter) interact using a Zoom audio channel with the trials being audio recorded for later analysis. Audio re-direction² is used on the server hosting the game so that verbalized text-to-speech (TTS) response are played out to the microphone on the server so fellow players on the Zoom call could hear the Wizard's response. The DesertWoZ environment logs all responses made by the Wizard including their utterance and when the utterance was made during a trial for later analysis.

In terms of the underlying language technology in use, our philosophy is to use the technology only up to the point that it enables participants to be deceived (i.e. believe they are interacting with an autonomous CUI and not a human confederate), as we are interested in understanding whether the responses envisioned result in team performance is comparable to a prior all human participant study [13]. Therefore, the sole language technology in use in the platform is Text to Speech (TTS). We use Mozilla's TTS³ engine to verbalize the Operator agent's response, which was chosen due to its open-source nature and good quality voices[1]. Since the number of responses the Operator agent could provide was finite and to eliminate any potential for performance bottlenecks, all possible responses were cached beforehand so the platform could directly play the response as needed.

4 ENVISIONED DEMO AT CSCW

At CSCW 2022, attendees can participate as a ground player in the DesertWoZ environment. In addition, screen sharing of the presenter will allow attendees to see how the DesertWoZ is used to control what the Operator agent says.

¹Unity Technologies, CA, version 2018.4 LTS, <https://unity.com>

²VB-Cable Virtual Audio Device, <https://vb-audio.com/Cable/index.htm>

³TTS: Text-to-Speech for All, Mozilla Corporation, <https://github.com/mozilla/TTS>

5 DISCUSSION

DesertWoZ has been designed to specifically answer two key research questions we have set out in designing a collaborative CUI to support team-based online learning environments. First, we are interested in understanding how team performance and communication is affected when the human operator is replaced by a CUI with a limited set of responses. Second, we are interested in understanding whether an Operator conversational agent necessarily needs to listen to what the players are saying and respond to their inquiries or if an agent that simply "orders," ground players around is sufficient. We are also interested in collecting conversational data useful to training future team based CUIs.

Although this demo represents to our knowledge the first WoZ system to support CUIs for collaborative virtual learning environments, we foresee several future directions interested researchers may want to pursue. First, only a single experimenter could play the role of Wizard at a time. However, as the state of AI advances, it is quite likely teams may be composed of several AI and human agents working together to solve a common problem. Hence, extending such a platform to enable multiple CUIs to be controlled by one or more human Wizards may be valuable. Future work will also endeavor to create an interface for a Wizard playing other autonomous agent roles such as that of a ground player agent who can speak with the operator and other ground players.

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